

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

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Part – A Theory

5a https://www.w3schools.com/python/pandas/ref_df_filter.asp

5b https://www.w3schools.com/python/module_statistics.asp

5c <https://www.geeksforgeeks.org/python/creating-a-pandas-series-from-dictionary/>

5d <https://www.geeksforgeeks.org/pandas/boolean-indexing-in-pandas/>

PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame objects.

Code:

```
File Edit Format Run Options Window Help
import pandas as pd

data = {
    "name": ["Sally", "Mary", "John"],
    "age": [50, 40, 30],
    "qualified": [True, False, False]
}
df = pd.DataFrame(data)

print(df)
```

Object:

```
= RESTART: C:/Users/nidhi/Downloads/N
.py
   name  age  qualified
0  Sally   50        True
1  Mary   40        False
2  John   30        False
```

5b. Write a python code to show statistical information on a given data set.

Code:

```
# Import statistics Library
import statistics
```

```
#Calculate the high middle values
print("---Calculate the high middle values---")
print(statistics.median_high([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median_high([1, 3, 5, 7, 9, 11]))
print(statistics.median_high([-11, 5.5, -3.4, 7.1, -9, 22]))
```

```
# Calculate harmonic mean
print("---Calculate harmonic mean---")
print(statistics.harmonic_mean([40, 60, 80]))
print(statistics.harmonic_mean([10, 30, 50, 70, 90]))
```

```
# Calculate the standard deviation from an entire population
print("---Calculate the standard deviation from an entire population---")
print(statistics.pstdev([1, 3, 5, 7, 9, 11]))
print(statistics.pstdev([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.pstdev([-11, 5.5, -3.4, 7.1]))
print(statistics.pstdev([1, 30, 50, 100]))
```

```
# Calculate the variance from a sample of data
print("---# Calculate the variance from a sample of data---")
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print(statistics.variance([1, 30, 50, 100]))
```

```
# Calculate the variance of an entire population
print("---# Calculate the variance of an entire population---")
print(statistics.pvariance([1, 3, 5, 7, 9, 11]))
```

```
print(statistics.pvariance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.pvariance([-11, 5.5, -3.4, 7.1]))
print(statistics.pvariance([1, 30, 50, 100]))
```

output:

```

---Calculate the high middle values---
7
7
5.5
---Calculate harmonic mean---
55.38461538461538
27.97513321492007
---Calculate the standard deviation from an entire population---
3.415650255319866
0.6322358912796886
7.287146217827662
36.09276243237694
---# Calculate the variance from a sample of data---
14
0.47966666666666663
70.80333333333333
1736.9166666666667
---# Calculate the variance of an entire population---
11.666666666666666
0.3997222222222222
53.1025
1302.6875

```

5c. Write a python code to create pandas series from dictionaries.

Code:

```
# import the pandas lib as pd
import pandas as pd

# create a dictionary
dictionary = {'A': 10, 'B': 20, 'C': 30}

# create a series
series = pd.Series(dictionary)

print(series)
```

output:

```
.py
A      10
B      20
C      30
dtype: int64
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

Code:

```
File Edit Format View Options Window Help
# importing pandas as pd
import pandas as pd

# dictionary of lists
dict = {'name':["aparna", "pankaj", "sudhir", "Geeku"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[90, 40, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)
```

output:

```
.py
      name  degree  score
True  aparna    MBA    90
False  pankaj    BCA    40
True   sudhir  M.Tech    80
False  Geeku    MBA    98
|
```